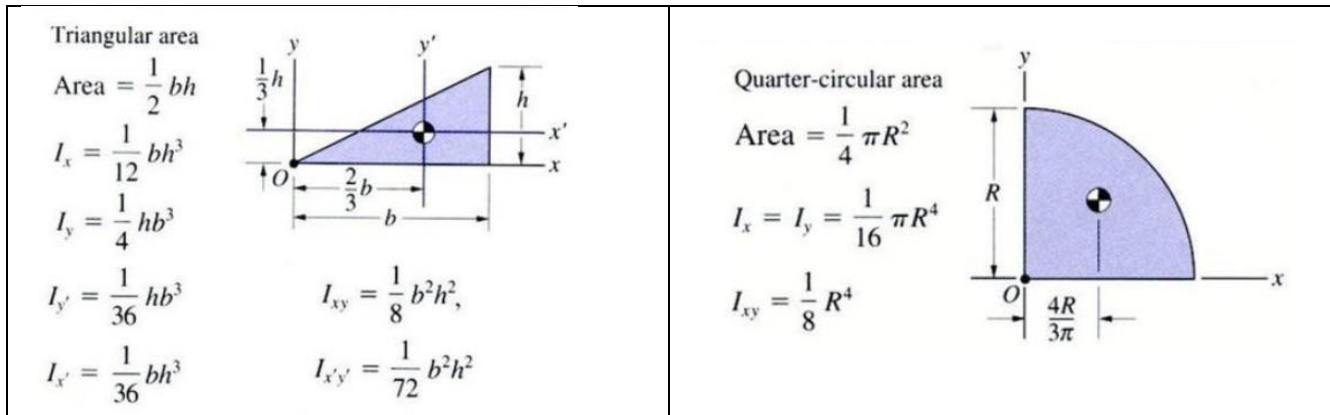


Hibbeler's Official Crib Sheet: Product of Inertia (POI or I_{xy}) for Right Triangles or Quarter Circles

Hibbeler doesn't tell the whole story with his I_{xy} formulas for right triangles or quarter circles. This is dangerous for our students; they will not realize what Hibbeler is not saying.

For right triangles, he doesn't note that the I_{xy} or $I_{x'y'}$ signs change depending on the orientation of the shape.

For quarter circles, he doesn't give a centroidal ($I_{x'y'}$) formula at all, and he doesn't note that the sign will change on I_{xy} depending on the quadrant. So if students ever have a quarter circle that doesn't touch both the x and y axes, then they're out of luck; no $I_{x'y'}$ formula is given.



This is how I give the right triangle and quarter circular I_{xy} formulas to my students (and then they have to use the $I_{xy} = I_{bar} + A d_x d_y$ formula to get the I_{xy} about any other axes other than the centroidal axes.

